

06 — Metadata forensics

What it detects: signs a PDF was edited, re-exported or assembled after it was created: editing software fingerprints, inconsistent or missing timestamps, incremental-save revisions, XMP edit history that contradicts the file, orphaned objects, embedded JavaScript, and a stripped metadata block.

Code: `services/forensics/metadata_forensics.py (analyze_pdf_metadata)`, run by `tasks/metadata_forensics_task.py (run_metadata_forensics)`; stores `check_type = metadata_forensics`.

What is read

Using **pikepdf** on the PDF's own object model (never a page render):

- the **Info dictionary** (`/Title /Author /Creator /Producer /CreationDate /ModDate`);
- the separate **XMP metadata stream**, including custom namespaces and the `xmpMM:History` revision chain;
- the **document ID chain** (`/ID`);
- the file's real **incremental-update / xref-revision structure** (`_walk_xref_chain`);
- **reachability**: which objects are unreachable from `/Root / /Info` (orphans);
- JavaScript / `OpenAction` , optional content groups, and digital-signature fields.

Findings

Every finding is `{finding, severity, description, data?}`. `severity` is `info | low | medium | high`. `info` findings are context (they show the raw metadata) and never affect the result.

finding	Severity	Meaning
<code>pdf_unreadable</code>	high	The file could not be opened as a PDF
<code>metadata_entirely_stripped</code>	high	No Info and no XMP metadata at all
<code>editing_software_detected</code>	high	Producer/Creator/ <code>softwareAgent</code> matches an image/graphics editor (<code>METADATA_FORENSICS_EDITING_SOFTWARE_NAMES</code> : photoshop, gimp, illustrator, affinity, coreldraw, paint.net, pixlr, canva, inkscape)
<code>pdf_editor_detected</code>	medium	Producer/Creator names an online or desktop PDF editor (<code>METADATA_FORENSICS_PDF_EDITOR_NAMES</code> : iLovePDF, Smallpdf, Sejda, PDFescape, PDF-XChange Editor, PhantomPDF / Foxit PDF Editor, Nitro Pro, PDFelement, Soda PDF, pdfFiller, DocHub, PDF Candy). One finding per tool. Print drivers of the same vendors are not listed
<code>mod_date_after_creation_date</code>	high	ModDate later than CreationDate by more than <code>METADATA_FORENSICS_MOD_DATE_THRESHOLD_SECONDS</code> (default 1 s — only jitter tolerance; being modified at all is the anomaly)
<code>info_xmp_creation_date_mismatch</code>	high	Info and XMP creation dates differ by > 1 minute

finding	Severity	Meaning
info_xmp_mod_date_mismatch	high	Info and XMP modification dates differ by > 1 minute
info_xmp_producer_mismatch	medium	Info /Producer ≠ XMP producer
incremental_updates_present	medium (1 update) / high (> 1)	The file has been saved incrementally
history_editing_tool_anomaly	medium	An XMP history entry's softwareAgent names an image editor or a PDF editor
history_scanned_document_edited	high	An XMP history entry's action is editedScannedDoc — Acrobat's "Edit scanned document": the scan was converted to editable text and edited
history_edit_after_final_date	high	An XMP history event is later than the file's modified date (+1 min)
history_shorter_than_revisions	medium	Fewer history entries than xref revisions
history_absent_despite_revisions	medium	Revisions exist but the history is empty
orphaned_objects	medium (orphans found) / info	Content objects present but unreachable from /Root or /Info — typical of deleted content. Only content counts: streams, fonts, images and form XObjects, pages, annotations (data.kinds). A bare array, number or plain dictionary (macOS/iOS Quartz leaves its page box [0 0 612 792]), or an empty stream (/Length ≤ 16 or nothing once decoded — PDFium leaves an empty Form XObject per page) is not counted (data.trivial_ignored)
javascript_or_openaction	medium (flagged) / info	Embedded JS or auto-run action
optional_content_groups	low / info	Layers present

finding	Severity	Meaning
editable_text_over_scan	medium	Page structure (<code>services/forensics/page_structure.py</code> , from the drawing order): an image covering $\geq 90\%$ of the page drawn before visible live text (≥ 15 runs spanning $\geq 50\%$ of the page height) — a scan converted to editable text, whatever the fonts are called (a converter's <code>-NNNN</code> subsets, listed in <code>data.converter_font_pages</code> , or ordinary <code>ABCDEF+</code> subsets as Acrobat's editor leaves them). Does not depend on XMP, so it still holds when the metadata is stripped. An image drawn <i>over</i> the text (a flattened page pasted on top) is not this structure. A normal searchable scan keeps its OCR text invisible (render mode 3). Such pages also get the ghost-content check (07a)
producer_scrubbed	medium	The XMP packet was written by Adobe's own library (<code>x:xmptk</code> "Adobe XMP Core ...", read from the raw stream), yet Info Producer/Creator, XMP Producer/CreatorTool and the edit history are all empty. Adobe software always records its name and saves there, so they were removed afterwards. A fully empty file is <code>metadata_entirely_stripped</code> instead
web_page_origin	low / medium	The producer is a browser's or an HTML engine's PDF output (PDFium, Skia/PDF, Chrome, wkhtmltopdf, Puppeteer, Playwright, WeasyPrint, Prince...): the document was built as a web page and printed. Medium (<code>metadata.web_page_origin</code> , 10) when it also has no CreationDate or uses ≥ 3 stock colours of a CSS framework (Bootstrap 5, Tailwind — <code>services/forensics/pdf_facts.py</code>); low (shown only) with the engine alone, since school portals print from browsers too
digital_signature	info	Signature field present / signed / unsigned / absent
info_dictionary, xmp_metadata, xmp_history, revision_count, document_id_chain	info	Raw context

The check `result` is **flag** if any finding is **medium** or **high**, else **pass** .

All `metadata.*` rules together add at most the company's metadata cap (default 40) to a case's score — see 11.

Output shape (real, truncated)

```
{
  "result": "flag",
  "details": [
    {"finding": "orphaned_objects", "severity": "medium",
      "description": "Found 25 object(s) present in the file but not reachable from /Root or /Info – likely left...",
      "data": {"count": 25, "object_numbers": [2, 3, "..."]}},
    {"finding": "info_dictionary", "severity": "info", "description": "Info dictionary has 6 field(s).",
      "data": {"/Creator": "Adobe Photoshop 2024 (Windows)", "/ModDate": "D:20260902164500Z",
        "/CreationDate": "D:20260801090000Z", "/Producer": "Adobe Photoshop 2024 (Windows)", "...":
        "..."}
    ]
  }
```

`details` is a **list** of findings (the same shape ELA, copy-move, duplicate and visual review use, so one findings-list UI renders them all). These findings carry no `page` / `bounding_box` ; metadata is not positional.

Known limitations

- **PDF only.** Images and Word files are not analyzed.
- Many legitimate tools leave incremental updates or editor fingerprints (a scanned-then-annotated invoice, for example). A finding is a reason to look, not proof.
- A determined forger can rewrite or strip metadata; `metadata_entirely_stripped` is itself scored because clean-room generators rarely strip everything.
- Uploads are PDF-only, so every new document is analysed; a non-PDF stored before that rule is a quiet no-op.
- The visual-review task cross-references this result (`metadata_forensics_correlation`) to strengthen or explain its own findings; that cross-reference is deliberately **not scored** to avoid double counting.

Risk rules fed

rule_id	Finding	Fires on	Weight	Severity
<code>metadata.pdf_unreadable</code>	<code>pdf_unreadable</code>	any	20	high
<code>metadata.stripped</code>	<code>metadata_entirely_stripped</code>	any	12	medium
<code>metadata.editing_software_detected</code>	<code>editing_software_detected</code>	any	30	high
<code>metadata.creation_date_mismatch</code>	<code>info_xmp_creation_date_mismatch</code>	any	18	high
<code>metadata.mod_date_mismatch</code>	<code>info_xmp_mod_date_mismatch</code>	any	18	high
<code>metadata.producer_mismatch</code>	<code>info_xmp_producer_mismatch</code>	any	10	medium
<code>metadata.pdf_editor_producer</code>	<code>pdf_editor_detected</code>	any	10	medium
<code>metadata.modified_after_creation</code>	<code>mod_date_after_creation_date</code>	any	15	high
<code>metadata.incremental_update_single</code>	<code>incremental_updates_present</code>	medium	8	medium
<code>metadata.incremental_update_multiple</code>	<code>incremental_updates_present</code>	high	20	high
<code>metadata.history_tool_anomaly</code>	<code>history_editing_tool_anomaly</code>	any	12	medium
<code>metadata.history_scanned_doc_edited</code>	<code>history_scanned_document_edited</code>	any	15	high
<code>metadata.editable_text_over_scan</code> (v2: any font naming)	<code>editable_text_over_scan</code>	medium, high	10	medium
<code>metadata.producer_scrubbed</code>	<code>producer_scrubbed</code>	any	10	medium

rule_id	Finding	Fires on	Weight	Severity
metadata.web_page_origin	web_page_origin	medium, high	10	medium
metadata.history_edit_after_final	history_edit_after_final_date	any	22	high
metadata.history_shorter_than_revisions	history_shorter_than_revisions	any	8	medium
metadata.history_absent_despite_revisions	history_absent_despite_revisions	any	8	medium
metadata.orphaned_objects (v3: content only, empty streams ignored)	orphaned_objects	medium, high	8	medium
metadata.javascript_or_openaction	javascript_or_openaction	medium, high	10	medium

Note the two severities: a rule's `severity` (how it is displayed) is independent of the *finding's* severity (which the rule may filter on with `severity_in`).